

NAADI — listening to the pulse of the plant

Cut unplanned pump downtime **35–45%** in Indian chemical plants with a vendor-neutral, plant-owned vibration-intelligence stack.

Continuous vibration sensing and interpretable machine learning on 25 bad-actor pumps: ~₹37 lakh to install, ~11-month payback, prototype already validated on a public bearing benchmark.

Problem statement (Annexure): AI-driven predictive maintenance to improve asset reliability and maintenance efficiency in the chemicals & petrochemicals sector.

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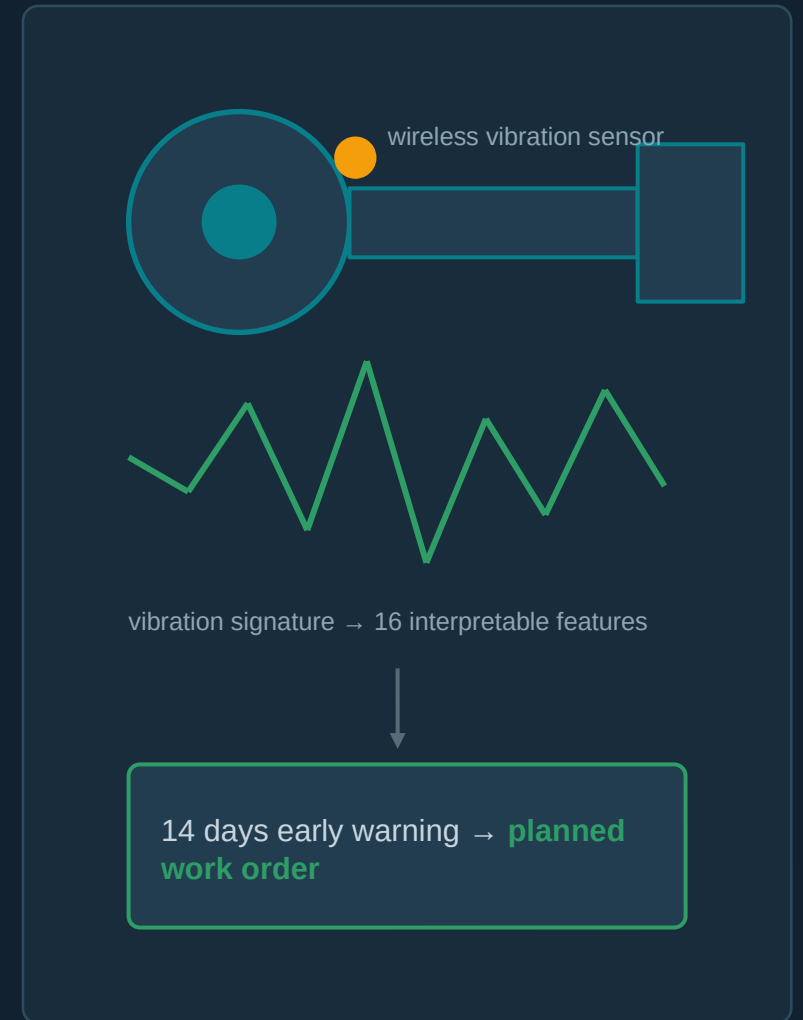
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Twenty-five bad-actor pumps expose a mid-size plant to ~₹1.4 crore a year because maintenance runs on the calendar, not on machine condition

60 h/yr

unplanned outage caused by these pumps (est.)

₹1.9 lakh/h

contribution margin lost while down (est.)

12 /yr

repair events across the 25 pumps (est.)

~₹1.4 Cr

annual exposure: lost margin + repairs (est.)

USE CASE & SYSTEM BOUNDARY

- Representative mid-size speciality-chemicals plant: ₹500 Cr revenue, 8,000 operating h/yr, ~200 pumps.
- Boundary: the 25 critical API 610-class centrifugal pumps in continuous service whose trip stops production.
- Practice today: calendar-based preventive maintenance plus breakdown response; no continuous condition data.

WHY IT FAILS — ROOT CAUSES

- Failures strike at random: the foundational RCM study found 89% of failure modes have no wear-out age — fixed-interval PM cannot anticipate them [4].
- Bearing and seal degradation dominates pump failures [5], and develops over days-to-weeks — invisible between monthly rounds.
- Repair only after functional failure → secondary damage, emergency spares, multi-hour restarts.

The problem we solve: detect developing faults on these 25 pumps at least two weeks before functional failure, and convert every detection into a planned work order — without replacing any existing plant system.

Continuous monitoring is proven worldwide — but OEM cloud lock-in prices out mid-size Indian plants: the gap is an **open, plant-owned stack**

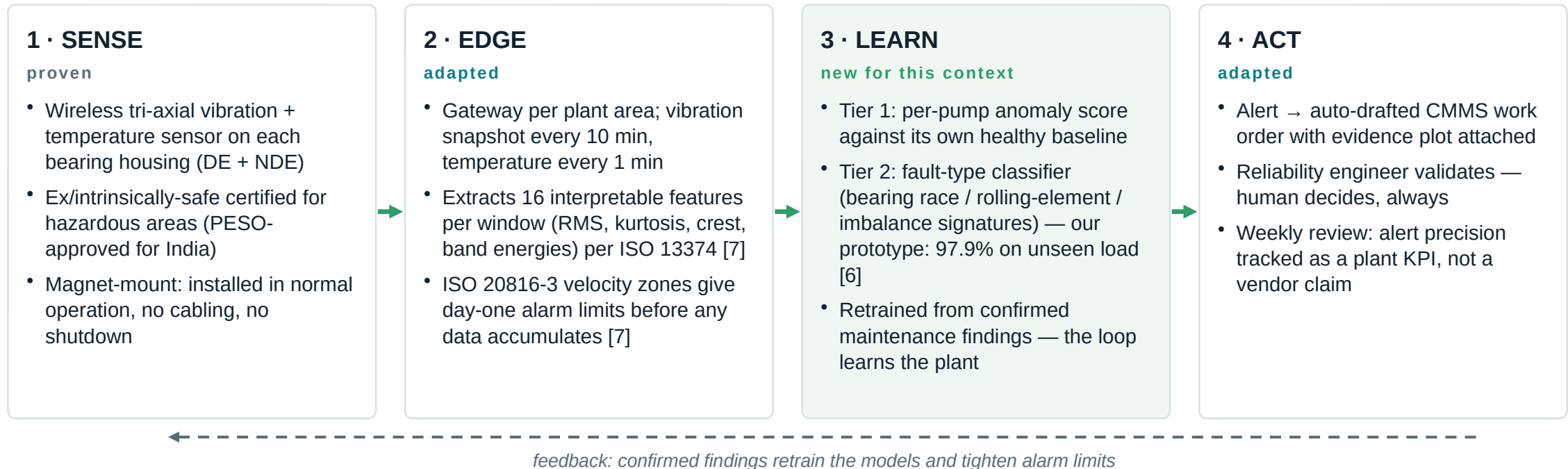
Approach	Detection lead time (30)	Total cost of ownership (20)	Skill dependence (10)	Data ownership (10)	Retrofit ease (10)	MSME scalability (20)	Score / 5
Reactive (run-to-failure)	None	Poor — downtime pays	Low	—	—	Poor	2.3
Calendar preventive (Indian default)	None for random faults	Medium	Low	—	—	Medium	3.0
Monthly offline vibration rounds	≤ route interval	Medium	High (analyst)	Plant	Good	Poor	2.9
OEM / proprietary cloud monitoring	Days–weeks	High (per-point + subscription)	Low	Vendor silo	Medium	Poor	2.9
Open edge stack (proposed)	Days–weeks	Low–medium	Medium	Plant-owned	Good	Good — shared clusters	4.3

Weights in brackets (sum 100); scores 1–5 are team judgment from the sourced characteristics of each approach — full scoring note in our research log.

FIVE DESIGN PRINCIPLES THIS ANALYSIS FIXES

- Continuous, not periodic — no month-long blind windows
- Interpretable features a reliability engineer can audit — no black box
- Open protocols (OPC UA) and plant-owned data — no vendor silo
- Ex-certified wireless retrofit — no shutdown, no cabling
- Human-in-loop: every alert becomes a CMMS work order, never an automatic action

NAADI is a four-layer retrofit: **sense** → **edge** → **learn** → **act**, closing the loop from bearing vibration to a planned CMMS work order



WHY THIS IS DIFFERENT — THREE DEFENSIBLE POINTS

Openly validated: detection metrics published from a public benchmark with reproducible code — judge the evidence, not a brochure [6]. **Plant-owned:** open protocols, models and data stay with the plant — no subscription lock-in. **Priced for India:** ~₹1.5 lakh per pump all-in (est.), an architecture MSME clusters can share.

A prototype we trained on the public CWRU benchmark separates faulty from healthy bearings with **zero false alarms** — under leakage-free evaluation on an unseen load

MECHANISM — WHY VIBRATION WORKS

- A spalling bearing race strikes every rolling element at a defect-specific frequency; energy shifts into characteristic bands and the signal turns impulsive (kurtosis, crest factor rise).
- 16 classical indicators per 0.17 s window — every input auditable by a vibration analyst; a random forest maps them to fault classes.

Evaluation honesty: no random window splits (they leak near-duplicates). We hold out whole recordings and an entire unseen 3 hp load condition.

READINESS & INTEGRATION

- Sensors, gateways, OPC UA, CMMS APIs: mature industrial products (TRL 9); the integrated NAADI workflow is TRL 6–7 — exactly what the pilot derisks.
- Alarm limits from ISO 20816-3 zones on day one; learning takes over as data accumulates [7].

PROTOTYPE RESULTS — CWRU BENCHMARK [6]

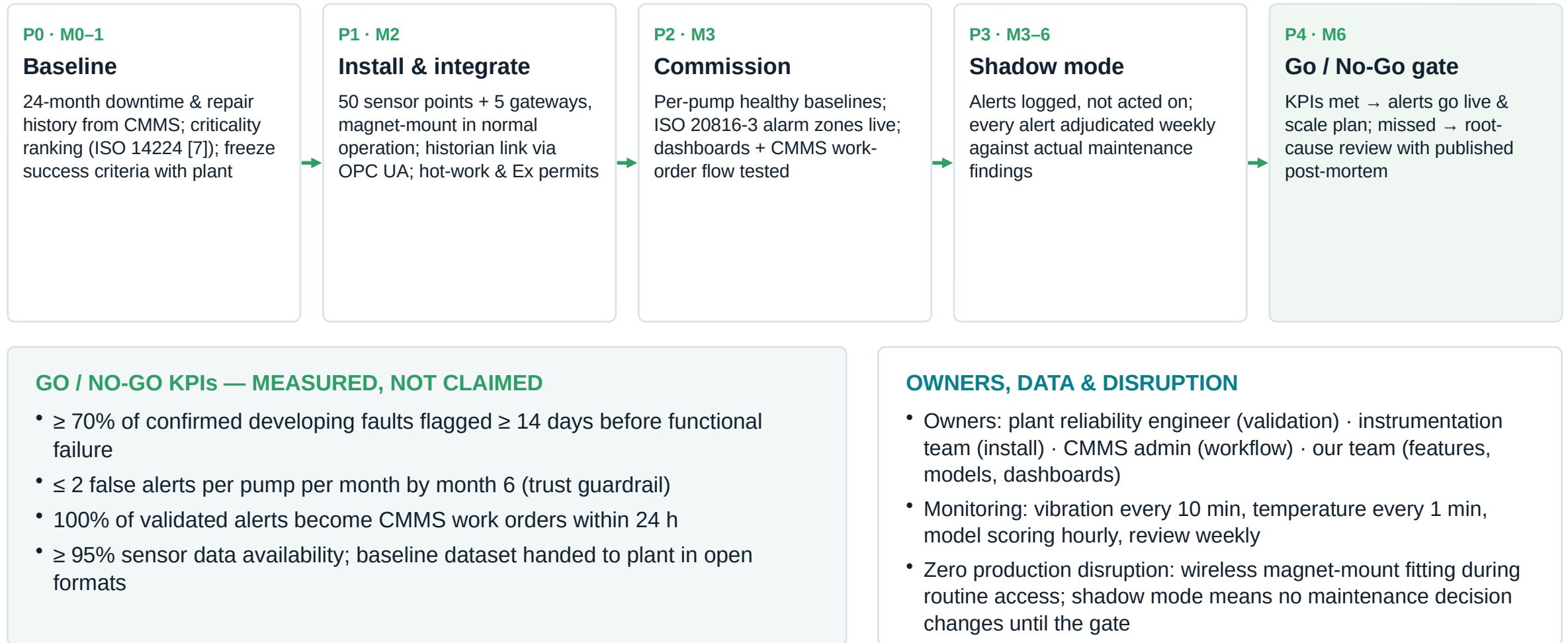
Metric (test = unseen 3 hp load)	Result	Protocol
Fault vs healthy — precision / recall	100% / 100%	1,123 windows
Healthy windows misflagged	0 of 237	false-alarm proxy
4-class fault-type accuracy	97.9%	fault-type mixups only
Grouped 5-fold CV (by recording)	98.9%	no recording crosses folds
Logistic baseline (same features)	97.2%	features carry the signal

64 recordings → 4,367 windows · 12 kHz drive-end accel. · code + metrics in our open repository

LIMITS WE ARE NOT HIDING

- CWRU is a clean test rig with seeded faults — plant noise, variable speed and real degradations will cut performance; the pilot KPI is set at $\geq 70\%$ detection, not 97%.
- Real failure examples are scarce at first → anomaly-first strategy; MEMS sensors trade bandwidth vs piezo — chosen for cost, validated in Phase 2.

A six-month, **shadow-mode pilot** on 25 pumps proves the value before a single maintenance decision depends on it



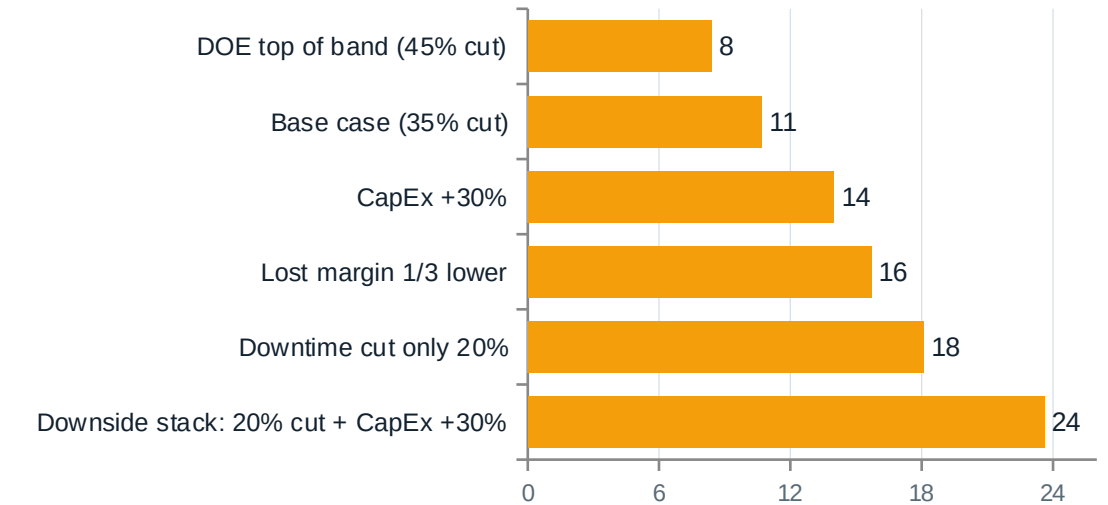
₹37 lakh installed, ~₹41 lakh net benefit a year: **payback in about 11 months**, and under 24 months even in the stacked downside

THE CALCULATION, NOT JUST THE ANSWER

Lost margin = ₹500Cr × 30% ÷ 8,000h = ₹1.9 L/h
 Downtime cut = 60h × 35% [3] × 1.9 = ₹39.4 L
 Repair save = 12 × ₹2.5L × 10% [3] = ₹ 3.0 L
 Catastrophic avoided = 2 × ₹2.5L = ₹ 5.0 L
Benefit ₹47.4L – OpEx ₹6L = ₹41.4 L/yr
CapEx ₹37L ÷ 41.4 × 12 ≈ 11 months
 5-yr NPV @12% ≈ ₹1.1 Cr · ₹0.64L per avoided hour

CapEx: 50 Ex-rated sensor points (₹17.5 L) + 5 gateways (₹7.5 L) + integration (₹10 L) + training (₹2 L). DOE FEMP ranges applied at the low end [3]; plant figures are labelled estimates — full live-formula workbook in our repository.

SENSITIVITY — PAYBACK IN MONTHS



ENVIRONMENT & SAFETY GAINS

- Seal failures are a leading release path [5]: early detection means fewer leak events, spills and emergency flaring/venting.
- Pumping ≈ 20% of world electric-motor demand [10]: the same condition data flags cavitation and off-BEP running for energy fixes.

TOP RISKS → SPECIFIC MITIGATIONS

- False alarms erode trust → shadow mode + ≤2/pump-month gate. Hazardous areas → PESO/IECEX intrinsically-safe sensors only.
- OT cybersecurity → one-way data flow, segregated network. Skills → weekly analyst review + operator training in CapEx.

Benchmark-validated today, plant-validated in six months — then an **open reliability stack** India's chemical clusters can share

VALIDATED ALREADY (SOURCED / COMPUTED)

- Effectiveness band: 35–45% downtime cut for working PdM programmes — US DOE FEMP [3]
- Failure logic: 89% of failure modes not age-related; bearings/seals dominate pumps [4][5]
- Detection: our prototype, leakage-free evaluation on public CWRU benchmark — 97.9% unseen-load accuracy, 0 false alarms [6]
- Standards path exists: ISO 20816-3 · 13374 · 17359 · 14224 [7]

STILL NEEDS PILOT PROOF (WE SAID SO ON EVERY SLIDE)

- Detection lead time and false-alert rate under real plant noise and variable operation
- The 60 h/yr baseline and ₹1.9 L/h margin — replaced by the host plant's actual history in P0
- Indian installed cost points (sensor quotes, Ex certification, integration effort)
- Model transfer from benchmark features to plant-specific baselines

M0–6 Pilot

25 pumps, one plant · shadow → live at the gate



M6–12 Plant-wide

100+ assets: compressors, agitators, fans · plant reliability dashboard



M12–24 Cluster scale

Shared-service model for MSME chemical clusters · aligned with SAMARTH Udyog centres & NM-ICPS hubs [8][9]

Our ask of industry — one pilot partner: 25 critical pumps, 12 months of CMMS & historian history, and site access for 6 months. We bring sensors, models, dashboards and open code — the plant keeps the data, the models and the results, published either way.

If the jury remembers one line:

Listen to the machines before they stop the plant — ₹37 lakh, 11-month payback, and the plant keeps the brains.